Comprehensive Applied Mathematics & Engineering Handbook

A master reference compendium containing core definitions, identities, and worked model solutions.

1. Differentiation & Differential Calculus

Fundamental Rules:

• Power Rule: d/dx(x^n) = n*x^(n-1)

• Product Rule: d/dx[u*v] = u'v + uv'

• Quotient Rule: d/dx[u/v] = (u'v - uv') / v^2

• Chain Rule: d/dx[f(g(x))] = f'(g(x)) * g'(x)

• Derivatives of Transcendentals: d/dx(e^(ax)) = a*e^(ax), d/dx(ln x) = 1/x, d/dx(sin x) = cos x, d/dx(cos x) = -sin x, d/dx(tan x) = sec^2(x)

Model Problem: Find the derivative of f(x) = x^2 * sin(3x).

Worked Solution: Using product rule and chain rule:

f'(x) = d/dx(x^2) * sin(3x) + x^2 * d/dx(sin(3x)) = 2x*sin(3x) + x^2*(3*cos(3x)) = 2x*sin(3x) + 3x^2*cos(3x).

2. Integral Calculus

Standard Integrals:

• ∫ x^n dx = (x^(n+1))/(n+1) + C (for n ≠ -1); ∫ (1/x) dx = ln|x| + C

• ∫ e^(ax) dx = (1/a)*e^(ax) + C; ∫ sin(ax) dx = -(1/a)*cos(ax) + C; ∫ cos(ax) dx = (1/a)*sin(ax) + C

• Integration by Parts: ∫ u dv = u*v - ∫ v du

Model Problem: Evaluate ∫ x * e^(2x) dx.

Worked Solution: Let u = x (hence du = dx) and dv = e^(2x) dx (hence v = (1/2)*e^(2x)).

∫ x*e^(2x) dx = x * ((1/2)*e^(2x)) - ∫ (1/2)*e^(2x) dx = (1/2)*x*e^(2x) - (1/4)*e^(2x) + C = (1/4)*e^(2x)*(2x - 1) + C.

3. Laplace Transforms

Definition: L{f(t)} = F(s) = ∫[0 to ∞] e^(-st) * f(t) dt.

Standard Transform Pairs:

• L{1} = 1/s (s > 0)

• L{t^n} = n! / s^(n+1) (n = 1, 2, 3...)

• L{e^(at)} = 1 / (s - a)

• L{sin(at)} = a / (s^2 + a^2); L{cos(at)} = s / (s^2 + a^2)

• First Shifting Theorem: L{e^(at) * f(t)} = F(s - a)

• Transform of Derivatives: L{y'(t)} = s*Y(s) - y(0); L{y''(t)} = s^2*Y(s) - s*y(0) - y'(0)

Model Problem: Find the Laplace transform of f(t) = e^(3t) * cos(4t).

Worked Solution: Since L{cos(4t)} = s / (s^2 + 16), by applying the first shifting theorem (replace s with s - 3):

L{e^(3t) * cos(4t)} = (s - 3) / ((s - 3)^2 + 16) = (s - 3) / (s^2 - 6s + 25).

4. Differential Equations

First-Order Linear Differential Equation: dy/dx + P(x)*y = Q(x)

Integrating Factor: I(x) = exp(∫ P(x) dx). General Solution: y * I(x) = ∫ [Q(x) * I(x)] dx + C.

Second-Order Homogeneous DE with Constant Coefficients: a*y'' + b*y' + c*y = 0.

Characteristic Equation: a*r^2 + b*r + c = 0.

• Distinct real roots (r1, r2): y(x) = c1*e^(r1*x) + c2*e^(r2*x)

• Equal roots (r1 = r2 = r): y(x) = (c1 + c2*x)*e^(r*x)

• Complex roots (α ± iβ): y(x) = e^(α*x) * [c1*cos(βx) + c2*sin(βx)]

Model Problem: Solve y'' - 5y' + 6y = 0.

Worked Solution: Characteristic equation: r^2 - 5r + 6 = 0 => (r - 2)(r - 3) = 0.

Roots are r1 = 2, r2 = 3. General Solution: y(x) = c1*e^(2x) + c2*e^(3x).

5. Linear Algebra, Matrices & Complex Numbers

Determinant: For A = [[a, b], [c, d]], det(A) = ad - bc.
Eigenvalues & Eigenvectors: Solved using det(A - λI) = 0.
Complex Numbers: z = a + ib = r*(cos θ + i*sin θ) = r*e^(iθ). Modulus: |z| = √(a^2 + b^2).
Model Problem: Find eigenvalues of A = [[3, 1], [0, 2]].
Worked Solution: det([[3 - λ, 1], [0, 2 - λ]]) = (3 - λ)(2 - λ) - 0 = 0. Eigenvalues are λ1 = 3, λ2 = 2.